

[Linkden](https://www.linkedin.com/in/danishshahzad17) : www.linkedin.com/in/danishshahzad17

AGENTIC AI ENGINEER ROADMAP

By [Danish Shahzad](#)

[YouTube](https://www.youtube.com/@danishshahzadai) : [http://www.youtube.com/@danishshahzadai](https://www.youtube.com/@danishshahzadai)

[Linkden](https://www.linkedin.com/in/danishshahzad17) : [http://www.linkedin.com/in/danishshahzad17](https://www.linkedin.com/in/danishshahzad17)

STUDY 1 – Foundation Python

Python Basics

- Variables, data types
- Local and global variables
- Lists, dictionaries, sets, tuples
- Arrays
- Operators
- Modules and packages
- User input
- Conditionals
- Loops (for / while)
- Functions
- Scope, return, args, kwargs
- Lambda & anonymous functions
- String methods
- File handling
- Virtual environments
- Using external libraries

OOP in Python

- OOP concepts
- Classes & objects
- Types of variables
- Types of methods
- Inner classes
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction
- Exception handling and Custom exceptions

[YouTube](https://www.youtube.com/@danishshahzadai): www.youtube.com/@danishshahzadai

[Linkden](https://www.linkedin.com/in/danishshahzad17) : www.linkedin.com/in/danishshahzad17

Advanced Python

- Async & await
- Multithreading & multiprocessing
- Decorators
- Context managers
- Generators & iterators
- Metaclasses
- Design patterns
- Logging
- Type hinting
- Testing (pytest, unittest)
- Dataclasses & typing (dataclasses, TypedDict)
- Protocol & Abstract Base Classes
- Dependency injection patterns
- Environment variables (dotenv)
- Configuration management

STUDY 2 – Data Processing & Visualization

NumPy

- Arrays & dtypes
- Reshaping & slicing
- Broadcasting
- Arithmetic operations
- Vectorization
- Functions
- Filtering
- Saving & loading data

Pandas

- DataFrames & Series
- Reading CSV / Excel / JSON
- Missing value handling
- Data cleaning
- GroupBy operations
- Joins & merges
- Window functions
- Feature creation

[YouTube](https://www.youtube.com/@danishshahzadai): www.youtube.com/@danishshahzadai

[Linkden](https://www.linkedin.com/in/danishshahzad17) : www.linkedin.com/in/danishshahzad17

Visualization

- Matplotlib
- Seaborn
- Line, bar, scatter, histogram plots
- Distribution plots
- Correlation heatmaps
- Data storytelling

STUDY 3 – Math & Statistics for AI

Statistics

- Descriptive vs inferential statistics
- Continuous vs discrete data
- Nominal vs ordinal data
- Mean, median, mode
- Variance & standard deviation
- Correlation & covariance
- Probability basics
- Random variables
- Normal distribution
- Hypothesis testing (t-test, chi-square, ANOVA)
- Bayesian statistics
- Maximum likelihood estimation

Linear Algebra

- Scalars, vectors, matrices
- Matrix operations
- Eigenvalues & eigenvectors

Calculus

- Derivatives
- Gradients
- Optimization intuition
- Gradient-based optimization (Adam, RMSprop)

[YouTube](https://www.youtube.com/@danishshahzadai): www.youtube.com/@danishshahzadai

STUDY 4 – Machine Learning Foundations

ML Fundamentals

- ML problem framing
- Supervised vs unsupervised learning

Data Preprocessing

- Missing values
- Outlier detection
- Scaling & normalization
- Encoding techniques
- Feature engineering

Model Workflow

- Train-test split
- Cross-validation
- Model evaluation

Algorithms

- Linear regression
- Logistic regression (binary & multiclass)
- Decision tree
- Random forest
- Support Vector Machine
- KNN
- Naive Bayes
- K-Means clustering

Optimization & Persistence

- Bias vs variance
- Hyperparameter tuning
- L1 & L2 regularization
- Model persistence (Pickle, Joblib)
- Ensemble methods (XGBoost, LightGBM, CatBoost)
- Dimensionality reduction (PCA, t-SNE, UMAP)
- Time series forecasting basics
- Model interpretability (SHAP, LIME)

STUDY 5 – Deep Learning

Foundations

- Neurons
- Neural networks
- Activation functions
- Loss functions
- Gradient descent
- Backpropagation

Frameworks

- TensorFlow
- PyTorch
- Keras

Architectures & Performance

- CNN
- Transfer learning
- RNN
- LSTM
- GPU usage
- TensorBoard
- Handling imbalanced datasets

STUDY 6 – NLP & Transformers

NLP Basics

- NLP pipeline
- Tokenization
- Stop words
- Stemming & lemmatization
- Regex

Text Representation

- Bag of Words
- TF-IDF
- Word embeddings
- Word2Vec

[Linkden](https://www.linkedin.com/in/danishshahzad17) : www.linkedin.com/in/danishshahzad17

Modern NLP

- Transformers
- Attention mechanism
- BERT
- Hugging Face ecosystem

STUDY 7 – Foundations of LLMs

LLM Concepts

- Transformer architecture
- Context window
- Tokenization
- Sampling strategies
- LLM inference optimization (vLLM, TensorRT-LLM)
- Mixture of Experts (MoE)
- Quantization (GPTQ, AWQ)
- Streaming responses

Prompting

- Prompt engineering
- Context engineering

APIs & Capabilities

- OpenAI API
- Claude API
- Gemini API
- Tool calling
- Function calling
- Structured outputs

[YouTube](https://www.youtube.com/@danishshahzadai): www.youtube.com/@danishshahzadai

STUDY 8 – RAG (Retrieval Augmented Generation)

RAG Basics

- RAG architecture
- Chunking strategies
- Embedding generation
- Advanced chunking (semantic, recursive)
- Multi-query & hypothetical document embeddings
- Self-RAG & corrective RAG
- Knowledge graph RAG
- Offline vs online RAG
- Agentic RAG

Retrieval

- RAG retrievers
- Hybrid search
- Reranking
- ColBERT retriever
- Metadata filtering
- Query rewriting

Evaluation

- RAGAS
- End-to-end RAG evaluation
- Human evaluation loops
- Groundedness & faithfulness metrics

STUDY 9 – Vector Databases

Vector Stores

- Pinecone
- FAISS
- Qdrant
- Weaviate
- Chroma

Core Concepts

- Indexing
- Similarity search

[Linkden](https://www.linkedin.com/in/danishshahzad17) : www.linkedin.com/in/danishshahzad17

- Scaling strategies
- Vector database benchmarking
- Hybrid search (dense + sparse)
- Approximate nearest neighbors (HNSW, IVF)
- Backup & disaster recovery
- Upsert & delete operations

STUDY 10 – Agentic AI Foundations

Agent Concepts

- AI agent definition
- Observe–Plan–Act–Learn loop
- ReAct pattern
- Agentic search (Tavily, Perplexity APIs)
- Reflexion & self-critique patterns
- Tree of Thoughts (ToT) , Graph of Thoughts (GoT), Chain of Thought (CoT)

Memory & Reasoning

- Short-term memory
- Long-term memory
- Semantic memory
- Planning & research layers
- Tool-augmented reasoning

STUDY 11 – Agentic Frameworks & SDKs

Frameworks

- LangChain
- LangGraph
- LangSmith

SDKs & Multi-Agent

- OpenAI Agents SDK
- Claude SDK
- Agent collaboration
- Task decomposition

[YouTube](https://www.youtube.com/@danishshahzadai): www.youtube.com/@danishshahzadai

STUDY 12 – Model Context Protocol (MCP)

MCP Fundamentals

- MCP architecture
- MCP servers
- MCP tools

Integration & Standardization

- MCP + APIs
- MCP + databases
- MCP + RAG
- Cross-platform agent skills
- MCP client implementation
- MCP server development
- MCP security & authentication
- MCP transport layers (HTTP, WebSocket)

STUDY 13 – Spec-Driven & AI-Driven Development

Development Paradigms

- Spec-Driven Development (SDD)
- AI-Driven Development (AIDD)

Tools & Workflow

- Spec Kit
- SKILL.md
- Claude Code
- Gemini CLI
- Claude Code → OpenAI / Claude SDK

Agent Types

- General agents
- Custom agents

STUDY 14 – Multi-Agent Systems

Architecture

- Agent orchestration
- Role-based agents
- Supervisor-worker patterns

Coordination & Factories

- Collaboration
- Negotiation
- Conflict resolution
- Agent factory
- Skill composition

STUDY 15 – Evaluation, Safety & Control

Safety & Human Oversight

- Guardrails
- Policy enforcement
- Human-in-the-loop (HITL)
- Shadow mode
- LangSmith evaluation datasets
- DeepEval framework

Evaluation

- Agent testing
- Reliability metrics
- Red teaming basics

[Linkden](https://www.linkedin.com/in/danishshahzad17) : www.linkedin.com/in/danishshahzad17

STUDY 16 – Backend, Deployment & Cloud

Backend

- FastAPI
- Pydantic
- SQLAlchemy ORM

Containers & Orchestration

- Docker
- Multi-stage builds
- Kubernetes
- kubectl-ai
- kagent

Cloud

- AWS
- GCP
- Azure

STUDY 17 – Dapr & Cloud-Native AI

Dapr Core

- State management
- Pub/Sub
- Service invocation
- Durable workflows

Agent Systems

- Cloud-agnostic agents
- Workflow orchestration

[YouTube](https://www.youtube.com/@danishshahzadai): www.youtube.com/@danishshahzadai

STUDY 18 – Observability & MLOps

MLOps

- MLflow
- Experiment tracking
- Model versioning
- CI/CD for ML

Observability

- OpenTelemetry
- Logs
- Metrics
- Traces

STUDY 19 – Monetization & Production Strategy

Business & Production

- Digital FTE
- Agent licensing
- Cost optimization
- ROI measurement
- Compliance & data privacy
- Security basics for AI systems